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The Enterprise AI Accountability Framework

Applying Four Decades of Telecommunications Accountability to the AI Transaction Economy

> Introducing the **Token Detail Record** (TDR), the **Proxy of Record** (POR), the **Model Routing Engine** (MRE), and the **Model Decision Record** (MDR) — and the maturity sequence that connects them.

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Executive Summary

Artificial intelligence has crossed the line from tool to infrastructure. It now mediates customer interactions, procurement, software delivery, research, and a growing population of autonomous workflows that act with minimal human supervision. The industry's attention remains fixed on capability — reasoning, agents, benchmarks, return on investment. Far less attention has been paid to the problem that will ultimately decide whether these deployments can be trusted, audited, governed, and paid for with confidence.

That problem is accountability, and it is not a new one.

The central claim of this paper is that **artificial intelligence is becoming a transaction economy**, and that transaction economies have a known life cycle. Every prompt, inference, retrieval, routing decision, and agent invocation is a metered economic event. As these events multiply, enterprises will need to independently record, attribute, govern, and reconcile them — exactly as the telecommunications industry learned to do across forty years of building the largest machine-generated transaction-accountability infrastructure ever assembled.

This paper argues that the AI industry should not reinvent that infrastructure. It should inherit it. To make the inheritance concrete, the paper coins four primitives, each minted directly from a proven telecom ancestor:

- **Token Detail Record (TDR)** — the atomic, per-event record of a single AI consumption charge, generated independently by *both sides*: the provider records what it intends to bill, the buyer records what it is willing to pay for, and the two are reconciled. *Heir to the Call Detail Record (CDR).*
- **Proxy of Record (POR)** — the enterprise-controlled system that observes, governs, and records AI activity. *Heir to the switch of record and the mediation layer.*
- **Model Routing Engine (MRE)** — the policy-driven system that selects the right model for a request. *Heir to least-cost routing (LCR).*
- **Model Decision Record (MDR)** — the record explaining why a given model was chosen. *Heir to routing-policy provenance — telecom's hardest-won and last-built discipline.*

The paper's second contribution is the observation that these four are not co-equal. They arrive in a **forced order**, each made necessary only when scale renders the previous one insufficient: *record consumption, then centralize control, then govern selection, then*

justify selection. Understanding that sequence tells an enterprise not only what to build, but when.

The AI Transaction Economy

Most organizations still file artificial intelligence under "software," and that filing is increasingly wrong. Conventional software is deterministic and its costs are bounded: inputs are processed, outputs are produced, and the marginal economic event is negligible. It emits logs and metrics, but it rarely creates a stream of independent, billable economic events.

AI does precisely that. Every interaction consumes metered resources, every model invocation carries a price, every retrieval and tool call adds to a running tab. The economy of AI is transactional at its root, and the simple request-and-response interaction of today's chat interface badly understates where it is heading. A single user-visible action will increasingly fan out into a cascade of underlying transactions — multiple models, multiple providers, agent-to-agent calls, vector retrievals, external APIs, and downstream business systems. What the user experiences as one question becomes dozens or hundreds of priced events, and agentic architectures will push those volumes up by orders of magnitude.

Each of those events is simultaneously a cost, a potential liability, and an accountability obligation. The question is no longer whether AI generates economic activity at scale. It is whether enterprises will be able to see, attribute, and verify that activity once it does — or whether they will be left holding a summarized invoice they cannot independently reconstruct.

What Telecommunications Already Solved

The telecom industry lived through exactly this transition. Early networks were simple: a call was set up, carried, torn down, and billed through a short and legible chain. Then the network became an ecosystem. Multiple carriers cooperated on a single call. Traffic crossed borders and regulatory regimes. Roaming, wholesale, interconnect, and settlement relationships multiplied. Transaction volumes exploded into the billions per day, and the industry discovered the uncomfortable truth at the center of every large transaction economy: **trust does not scale, and records do.**

The response was the **Call Detail Record (CDR)** — a standardized, per-event accounting record describing who originated a call, who received it, when, for how long, over which routes, and at what cost. The CDR was deceptively mundane and enormously consequential. Around it grew an entire discipline: mediation, rating, billing, revenue assurance, fraud management, interconnect settlement, dispute reconciliation, and audit. None of those functions could have existed without a trusted, interchangeable, per-event record beneath them.

The decisive property of the CDR was not its schema. It was that **both parties to a transaction generated one and reconciled against each other.** A carrier did not simply accept its partner's word for what crossed the boundary; it kept its own record and settled the difference. That two-sided discipline is what turned a log into an accounting primitive. Artificial intelligence today has no such record, on either side.

The Maturity Sequence

The temptation is to present a complete accountability stack and instruct enterprises to build all of it. That is not how telecom matured, and it is not how AI will. Accountability primitives arrive in a forced order, each driven into existence by a specific economic pressure the previous stage could no longer absorb:

> **Record consumption → centralize control → govern selection → justify selection.**

This sequence is the spine of the framework. The four primitives below are its named stages. An enterprise's position in the sequence tells it what is urgent now and what is premature — and the honest reality for nearly every organization today is that only the first stage is forced. The discipline is in resisting the later ones until scale actually demands them, and in recognizing the moment it does.

Stage One — The Token Detail Record (TDR)

Heir to the Call Detail Record.

The first pressure is cost, and cost forces accounting. The **Token Detail Record** is proposed as the atomic accounting unit of the AI economy: one record per consumption charge, capturing who consumed what, when, under which model and tariff, at what cost. Like the Call Detail Record it descends from, the TDR is a primitive that **both sides use** — the provider keeps its own TDRs to record what it intends to bill, and the buyer keeps its own, generated through its Proxy of Record, to record what it actually consumed and is willing to pay for. Neither is a copy of the other, and that is exactly the point: two independent accounts, reconciled against each other rather than one taken on faith. A candidate TDR carries the dimensions every downstream accountability function will need — transaction and correlation identifiers, timestamp, the tenant hierarchy (user, application, agent), provider and model, the full token vector (input, cached input, output, and reasoning tokens kept as non-overlapping buckets so nothing is double-counted), duration, region, and rated cost — together with the **tariff version the record was rated against**, so the charge can be deterministically recomputed in a dispute.

The purpose of the TDR is not billing. Billing is merely the first and most obvious use. The same record underwrites audit, chargeback, reconciliation, compliance, capacity planning, cost optimization, and settlement. Its significance lies not in its fields but in its role: it is the authoritative, independently held artifact each party reconciles against the other. Today the enterprise has only invoices, dashboards, and logs — all produced by the provider, none of them an independent, per-event record of what the buyer believes actually happened. That absence is the accountability gap, and the buyer-side TDR is the primitive that closes it.

Stage Two — The Proxy of Record (POR)

Heir to the switch of record and the mediation layer.

A record is only as trustworthy as the point that produces it, which raises the question that decides the entire framework: **who owns the record?** Today the answer is the provider. Enterprises depend on provider-generated invoices and dashboards as their primary source of truth — a posture that becomes untenable as spend grows and autonomous activity accelerates, because it leaves the buyer unable to independently verify the seller's bill.

The **Proxy of Record** is the enterprise-controlled point through which AI activity passes and at which the TDR is minted. It is the AI analogue of the telecom switch of record and the mediation layer that sat between the network and the billing system — trusted by neither party blindly, precisely because it belonged to the operator rather than to any single counterparty. Its responsibilities span request observation, policy enforcement, TDR generation, audit capture, and reconciliation support.

The POR is an architectural commitment, not a passive tap. To enforce policy and guarantee a complete record, it must sit in the request path as an active gateway — which makes it a control point with real availability and latency implications, and which is exactly why it must be engineered as infrastructure rather than bolted on as logging. The shift it represents is the same one telecom made: the enterprise stops consuming the provider's account of activity and starts keeping its own.

Stage Three — The Model Routing Engine (MRE)

Heir to least-cost routing.

The first two stages assume someone already chose the model. As model ecosystems grow from a handful of options into dozens, hundreds, or thousands — each trading off cost, latency, accuracy, model scale, context depth, geography, security, compliance, and contractual terms — that choice becomes an economic decision made thousands of times a day, and it can no longer be left to static assignment or developer habit.

The **Model Routing Engine** is the policy layer that evaluates the available models against enterprise objectives and selects the appropriate one for each request. It is the direct descendant of least-cost routing, the telecom engine that chose among interconnect partners on price, quality, and availability under operator policy. The MRE generalizes that idea well beyond cost. A mature engine weighs, at minimum:

- **Cost** — rated price per token bucket under the governing tariff.
- **Latency and availability** — responsiveness and current health of each candidate.
- **Accuracy and capability** — demonstrated quality for the task at hand.
- **Model scale** — parameter count and class, where size is a proxy for capability or for cost-efficiency depending on the workload.
- **Training scope** — what the model (or agent) was trained on, its domain coverage, recency, and the provenance and licensing of its training data.
- **Context depth** — the usable context window required by the request.
- **Geography** — the country and region in which inference runs and data resides, for sovereignty and residency obligations.
- **Security and compliance** — the model's security posture, certifications, and the regulatory constraints binding the request.
- **Contractual commitments** — vendor agreements, committed-spend tiers, and approved-vendor policy.

By scoring every candidate across these dimensions under enterprise policy, the MRE converts model selection from an ad hoc technical reflex into a governed business process.

A point of discipline belongs here. The MRE is a later stage for a reason: when an enterprise runs few models against well-understood workloads, static selection is the *correct* engineering choice, and a dynamic scorer is premature complexity. The MRE earns its place only when model count and workload variance exceed what fixed assignment can serve well. To matter as a primitive rather than as one more “LLM router,” the MRE must be defined by what distinguishes it from a plain dispatcher: it does not merely pick a model — it produces an auditable account of the pick.

Stage Four — The Model Decision Record (MDR)

Heir to routing-policy provenance — telecom’s last and hardest discipline.

If the TDR answers *what happened*, the MDR answers *why it happened*. It captures the decision that produced a model selection: the candidates available, the policy in force, the scores along each dimension — cost, latency, trust, compliance — the model chosen, the alternatives rejected, and the rationale. The MRE makes the decision; the MDR makes the decision accountable, closing the loop from policy to choice to consumption into a single auditable chain.

The MDR is deliberately last, and the reason is instructive. It is downstream of a live MRE that earlier stages do not require. More fundamentally, nothing *bills* on a decision rationale — there is no invoice forcing it into existence the way cost forced the TDR — so it is the primitive most easily deferred and most often missing. Telecommunications learned this the hard way: per-event billing records matured decades before routing-policy provenance became a first-class, auditable artifact, and the gap was felt every time an operator had to explain after the fact why traffic took the path it took. AI will rhyme. When costs spike, executives will ask why a more expensive model was used, what alternatives existed, and which policy drove the choice. The organizations that can answer will be the ones that treated the MDR as a primitive rather than an afterthought.

Reconciliation: Two Independent Records

It is essential to be precise about what the TDR is not. The framework does not propose a single, shared record co-owned by buyer and seller. It proposes the opposite, and the distinction is the whole point. **Each party keeps its own TDRs.** The provider generates its own to record what it intends to bill; the buyer’s Proxy of Record generates the buyer’s, an independent, authoritative account of what it consumed. Neither party accepts the other’s account on faith.

This is exactly how telecommunications worked, and why it worked. Each carrier generated its own Call Detail Records; settlement was not a matter of one carrier trusting the other’s bill, but of **reconciling two independently produced accounts** and resolving the difference. Trust was an output of reconciliation, not an input to it. The consumer’s TDR is the AI equivalent: the enterprise’s bible of what it is willing to pay for. When the provider’s invoice arrives, it is reconciled line-by-line against the POR’s own record rather than accepted as a summarized total. The enterprise that holds an independent record disputes from a position of evidence; the one that holds only the invoice disputes from a position of faith.

What remains to be built is not a shared record but the **standardization that makes independent reconciliation mechanical** — and here the telecom analogy carries a warning as much as a lesson. Telecom never fully standardized the CDR at the point of generation. Each switch vendor emitted its own format; an entire mediation industry existed precisely to normalize those vendor-specific records into something billable, and standardization was only ever achieved partially, at the interchange layer — constructs like the GSMA's TAP for roaming settlement — rather than at the switch. Reconciliation worked anyway, because each party held its own record and settled the difference, but it was harder and more artisanal than it needed to be. AI is heading for the same fragmentation: absent deliberate cooperation, every provider will emit its own TDR format and enterprises will build mediation to normalize them, exactly as telecom did.

But AI has an advantage telecom never had — foresight. The primitives can be named and built *early*, before the formats fragment and harden. If the industry agrees the base fields now — the irreducible core every TDR must carry — a strong working group, plausibly under an established standards body such as the ITU, can administer a **versioned standard** that evolves as the technology does. That is a higher order of cooperation than telecom ever achieved, and it is achievable only because the discipline is being defined while the economy is still young. The reconciliation era AI is entering will not be defined by the absence of trust — every high-volume transaction system, from telecom to payments to cloud, produces exceptions at scale that demand investigation. It will be defined by whether enterprises hold the independent records, and the industry agrees the standards, needed to resolve them.

Reference Architecture

The framework composes into a single path, with the maturity sequence layered onto it:

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        Applications and Agents
                   │
                   ▼
   Model Routing Engine (MRE) ──▶ Model Decision Record (MDR)    ← "why"
                   │
                   ▼
     Proxy of Record (POR) ───────▶ Token Detail Record (TDR)     ← buyer's TDR: "what"
                   │                               │
                   ▼                               ▼
           AI Providers ──▶ Provider TDRs / Invoice ──▶ RECONCILE
                                                       (buyer TDR vs. provider TDR / invo
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On the buyer's side the POR and MRE are the systems that emit the TDR and MDR; on the provider's side the provider generates its own TDRs and issues an invoice. Accountability is produced not by sharing a record but by **reconciling two independent ones** — the buyer's POR-generated TDR against the provider's TDRs and invoice — which is the substrate for audit, dispute, governance, and financial control.

Conclusion

The AI industry has moved through two phases — model creation, then enterprise adoption — and is entering a third defined not by intelligence but by accountability. The reason is structural: **AI cannot be fully commercialized until its financial transactions can be proven at the accounting level.** No mature market settles on faith. Until a buyer can independently demonstrate, from its own books, exactly what it consumed and what it should owe, AI spend remains an estimate rather than an audited liability — and an industry that bills in estimates has not finished becoming an industry.

This reframes where the competition ultimately goes. The current contest is fought on parameters, context windows, speed, and training scope — and those frontiers will eventually plateau and commoditize. The durable differentiator is operational: the ability to govern, attribute, reconcile, and prove the economic activity a model generates. That is a discipline a better model cannot resolve on its own, because it is not a capability of the model at all — it is a property of the system that operates it. The organizations that lead the third phase will not simply deploy the most capable models. They will own the most trusted framework for governing and verifying what those models do.

The contribution of this paper is twofold. First, a lexicon minted directly from telecommunications' proven primitives — the **Token Detail Record**, the **Proxy of Record**, the **Model Routing Engine**, and the **Model Decision Record** — so the AI industry can inherit four decades of accountability discipline rather than rediscover it. Second, the recognition that these primitives arrive in a forced order — **record, centralize, govern, justify** — so that enterprises know not only what to build but when each stage becomes necessary.

The call to action follows from the analogy that anchors the entire argument. A buyer's TDR is valuable the moment it exists — it is the independent bible for reconciling any invoice, and no standard is required to start keeping one. But telecom learned the limit of going it alone: because the CDR was never fully standardized at its source, reconciliation stayed harder than it should have been, mediated rather than mechanical. AI can do better, and only because it can act earlier. **The industry should agree the base fields of the Token Detail Record now** — not a record shared between buyer and seller, but a common, versioned schema for the TDRs each party keeps — and stand up the cooperative body, plausibly ITU-governed, to administer it before the formats fragment. Build the primitives early; agree the core now. The work of defining it should begin now, while it is still cheap to do.

**The future of artificial intelligence will not be defined solely by intelligence.
It will be defined, as every transaction economy before it, by accountability.**

About the author. Jerry W. Lambert, Sr. is Managing Director of SmarterDev.ai and a 25-year technology executive whose career spans the telecommunications accountability systems described in this paper — global billing and settlement at Pareteum, SIP telecom at Touchstone Systems, and the Excel Communications buildout — alongside enterprise IT modernization and AI transformation. He holds two U.S. patents in data processing. Contact: me@jerrywlambert.com | https://jerrywlambert.com